

Thanuish Kumar Sathiaraj

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M.Sc. Computer Science @ University of Stuttgart | Agentic AI & GenAI (LangGraph, MCP, Claude Code) | ex-Max Planck Institute |
Werkstudent, 20h/week

WORK EXPERIENCE

Student Research Assistant - Computer Vision

Nov 2025 - Apr 2026

Max Planck Institute for Intelligent Systems (Perceiving Systems, Data Team) - Tübingen, Germany

- Contributed to internal interface development for MAMA (Markerless Accurate Multi-person Motion Acquisition), managing containerized (Docker) backend services on Linux; work staged incrementally (2D to 3D to visualization) to ease debugging.
- Collaborated with PhD researchers in an Agile/Scrum setting on human body modelling pipelines, with version control (Git) and shared technical documentation.
- Supported database development and visualization of motion-capture outputs - organizing data pipelines, structuring outputs for analysis, and ensuring data consistency for downstream research use.

Machine Learning Engineer Intern

Apr 2024 - Sep 2024

Shiash Info-Solutions

- Developed an end-to-end convolutional neural network (CNN) image-classification pipeline in Python (TensorFlow/Keras), from data collection and cleaning through evaluation - an assigned project delivered independently under senior guidance.
- Applied data augmentation, normalization, and dropout regularization to reduce overfitting; optimized performance with Bayesian hyperparameter tuning for higher generalization accuracy.

PROJECTS

Market Sentinel - Multi-Agent Paper-Trading Research System (in development)

Jul 2026 - ongoing

Python, LangGraph, MCP (Model Context Protocol), yfinance, pandas, NumPy, pytest

- Designed and built a three-agent LangGraph pipeline for simulated trading research: a Watchdog computes deterministic technicals (RSI, MACD) over market data, an Evaluator runs a retrieval-augmented risk check (TF-IDF retrieval over a filings/news corpus plus live headlines, every decision citing its evidence), and an Executor places paper-broker orders - with the approval gate enforced as graph structure, not convention.
- Designed a deterministic execution guardrail: position sizing (fractional Kelly with hard risk caps) runs as pure, unit-tested Python that the LLM layer cannot alter, and is additionally exposed as an MCP (Model Context Protocol) server so external agent clients get the math without the model touching it.

JOBB HUNTT - AI-Assisted Job Application System

Jul 2026

HTML/CSS/JavaScript, File System Access API, SheetJS, docx.js, public job-board REST APIs (JSON)

- Specified an automation pipeline that parses job postings, extracts structured data, drafts tailored documents, exports DOCX, and logs outcomes to an Excel tracker; defined guardrails against fabricated claims and built a keyword-based ATS audit to benchmark output against raw job descriptions.
- Architected the pipeline for efficient model usage: the LLM's creative surface is reduced to one small structured JSON (every claim citing evidence ids from a canonical profile), while deterministic scripts handle validation, rendering, and coverage benchmarking - so smaller/cheaper models produce reliable, high-quality output inside the same guardrails; agent process definitions and reusable skill files (Claude Code skills) encode the workflow.

Sentiment Transfer in Movie Reviews - Research Project

2024 - 2025

Python, Hugging Face Transformers, T5, FLAN-T5, LLaMA 3, Sentence-Transformers

- Fine-tuned T5 and FLAN-T5 transformer models for supervised sentiment transfer; built a LLaMA 3 prompt-based in-context learning approach that achieved stronger polarity-flip performance under low-resource conditions.

Further projects: VisionARyMenu (Unity/ARCore AR app), Code Grabber (Python OCR + offline Whisper), Momentum (offline PWA)

SKILLS

Agentic AI & LLMs: LangGraph (multi-agent state machines), MCP (Model Context Protocol) servers, AI coding agents (Claude Code), LLM guardrails & evidence-cited generation, LLM Fine-tuning (T5, FLAN-T5, LLaMA 3), Prompt Engineering, Hugging Face Transformers

Programming: Python, C#, SQL, Bash, JavaScript, JSON, REST APIs

ML & Data: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, systematic evaluation & benchmarking

Tools & Testing: Git/GitHub (branching, PRs), Docker, Linux, pytest, NUnit, Jupyter

Spoken Languages: English (C1 - Proficient), German (A2 - currently studying)

EDUCATION

M.Sc. Computer Science

Apr 2025 - Jul 2027 (expected)

University of Stuttgart - Stuttgart, Germany

B.E. Computer Science & Engineering

Aug 2020 - Jun 2024

Anna University

INTERESTS

Dancing, Chess, Boxing, Boulderling, Hiking, Gaming, Travelling